The Benefits of POSIX®

LYNX SOFTWARE TECHNOLOGIES

The Benefits of POSIX® (Summary)

1. POSIX® Helps to Avoid Vendor Lock-in
2. POSIX® APIs are Valuable, Stable and Publicly Available
3. POSIX® Tutorials & Examples Can be Easily Found Online
4. POSIX® Increases the Number of Experienced Engineers that Can Work on Your Projects
5. POSIX® Reduces Total Cost of Ownership (TCO)
6. POSIX® Reduces Time-to-Market (TTM)

What Is POSIX®?

- The POSIX Standard is a document released by IEEE
- Describes a **P**ortable **O**perating **S**ystem **I**nterface based on UNI**X**
- Defines application programming interface (API), command line shells, and utility interfaces for software compatibility with variants of UNIX and other operating systems
- Popular and widely deployed
- Brings powerful set of features and capabilities for embedded application development

Following slides:

"What are the benefits of POSIX® to builders of complex safety- and security-critical embedded software systems?"

1. Avoid Vendor Lock-In

- Using any software API creates dependency
- Writing applications to proprietary APIs ties applications to vendor's operating system (OS)
- Makes changing vendors difficult or, in some cases, impossible
- Don't use a vendor's OS if not absolutely necessary
- Some questions to ask yourself when you start using any proprietary APIs are:
  - Is the OS suitable for the real-time needs of your embedded project?
  - Does the OS support the next CPU architecture you want to use?
  - Worst-case: Falling-out with your vendor once they've locked you into their proprietary APIs...

^These scenarios are not nearly as uncommon as they should be!

2. APIs are Valuable, Stable and Publicly Available

- IEEE standard published by The Open Group and available online
- No need to rely on proprietary documentation from a single-source vendor
- Expertly designed and widely standardized to promote portability.
- Port application from OS to OS, system to system, and project to project—all while maintaining the minimum amount of rework required.
- Well-designed (origins date back to birth C at Bell Labs) and has remained stable over time

3. Tutorials & Examples Can be Easily Found Online

- POSIX®-based OSs are by far the most popular and widely deployed
- Online tutorials, examples, and explanations of when you might want to use certain APIs are a boon to engineers that are not only new to POSIX® but to those familiar with POSIX® as well
- Examples:
  - How to fork-and-exec a process to run a new application
  - How to use pthread_create() to create a new thread of execution in a multi-threaded application
  - How to create and use a timer for an application's heartbeat
  - Etc.

4. More Experienced Engineers for Projects

- Engineering experience is a major pain point for builders of complex embedded safety- and security-critical software systems
- Linux is based on POSIX®
  - POSIX® application development experience translates from one POSIX® family OS to another
  - Lots of recent graduates and university students have experience with Linux
  - Large pool of experienced engineers that already program for it
- Don't use only a few experienced engineers that can use a given set of proprietary operating system APIs
- Don't wait while your new engineers learn those APIs, either

When you use POSIX® to build your applications, you gain the ability to leverage both existing and new engineering talent faster

5. Reduces Total Cost of Ownership (TCO)

- Lower TCO by maintaining leverage over your vendor (switching FROM proprietary):
  - If you can avoid vendor lock-in by using POSIX®, then you are not beholden to an OS vendor for your next project
  - If you had written your application with their proprietary APIs, then the OS vendor has the leverage—not you

- Lower TCO by leveraging portability (switching TO proprietary):
  - Many organizations underestimate the hidden cost of open source software, but **open source software is anything but free**
  - If you find that the TCO of an open source OS is too high, then having written your applications with POSIX® will enable you to more easily switch to a proprietary OS (where the support costs are built into the business model)

6. Reduces Time-to-Market (TTM)

- A proven set of APIs speeds development, reduces engineering costs
- If an engineer gets stuck on a problem, can always turn to a vast pool of available resources to help solve the problem
- Staff your project more easily and in a shorter amount of time (start your project sooner—and perhaps with less expensive engineers)

A greater degree of freedom and flexibility to switch OSs and vendors at your discretion, free and easily accessible documentation, a vast knowledge base, boosted engineering expertise, and reduced TTM & TCO are why POSIX® is so useful to builders of large and complex safety- and security-critical embedded systems.

Thank You